Research Paper Improvement Report

AI Event Photo Discovery Platform — research-inspired hierarchical face retrieval

Summary: The project extends the paper's structured identity-code retrieval idea into a practical event-photo system with adaptive indexing, multi-selfie fusion, quality-aware ranking, privacy controls, and measurable retrieval analytics.

1. Alignment with the research paper

The paper motivates structured identity representations and hierarchical search to reduce the cost of comparing a query against every identity. This project preserves that central idea while adapting it to event-photo retrieval.

Paper concept	Implemented project feature
Face embeddings	InsightFace generates 512-dimensional embeddings for detected faces.
Structured identity codes	Hierarchical clustering assigns codes such as **0-2-1** to indexed faces.
Hierarchical retrieval	The selfie first selects promising branches; Qdrant then searches only those branches.
Reduced search cost	The system measures indexed faces, candidate faces, branches, latency, and reduction percentage.

Academic accuracy: This is a retrieval-stage adaptation, not a complete reproduction of the paper's model-training procedure. The original paper trains a recognition backbone to predict identity-code tokens; this project uses a pretrained face model and applies structured indexing during retrieval.

2. Improvements implemented beyond the paper's core idea

Adaptive hierarchical indexing

• Hierarchy parameters are selected according to event size rather than using one fixed configuration.

• Small events use a smaller tree, while larger events use more branches and greater depth.

• Branching factor and maximum depth are returned as measurable search metadata.

Multi-selfie embedding fusion

• Attendees can upload multiple selfies in one request.

• Each valid face embedding is weighted by face-detection confidence.

• The weighted embeddings are normalized and fused into one query representation, making matching more robust to pose and lighting differences.

Quality-aware re-ranking

• Photo processing calculates a face-quality score using Laplacian sharpness variance.

• Quality is stored with the vector payload and used during result ranking.

• Final confidence combines 85% face similarity and 15% image quality, so blurry or low-quality detections are ranked lower when similarity is comparable.

Privacy-preserving biometric management

• Attendee selfies are decoded and processed in memory only.

• Raw selfies are never saved to the uploads directory, PostgreSQL, or Qdrant.

• Temporary upload streams are explicitly closed after processing.

• The biometric-data endpoint reports that no persistent attendee selfie data exists, avoiding deletion of shared event embeddings by mistake.

3. Measured project results

91.3% candidate reduction	**~140 ms** observed retrieval latency	**24** photos processed	**582** faces detected	**677** indexed embeddings	**55** hierarchy branches

These figures demonstrate the project's efficiency-oriented design. The exact values are workload-dependent and should be re-measured on the final evaluation dataset.

4. End-to-end system contribution

• Hosts upload event photographs.

• Celery processes images asynchronously and detects faces.

• InsightFace creates embeddings; quality scores are calculated.

• Qdrant stores vectors and structured hierarchy metadata.

• Attendees join through an event invite and submit one or more selfies.

• The fused query selects hierarchy branches and retrieves candidate faces.

• Results are quality-aware reranked, deduplicated by photo, and shown with research metrics.

5. Recommended evaluation for the paper/report

System	Purpose	Metrics
Flat search	Compare against every indexed face	Precision, recall, F1, latency
Fixed hierarchy	Measure the original project approach	Candidate reduction, recall, latency
Proposed system	Evaluate adaptive indexing, fusion, and reranking	Precision, recall, F1, latency, quality impact

The strongest research claim should be based on measured results: the proposed system should reduce candidate comparisons while maintaining or improving recall compared with flat retrieval.

6. Final contribution statement

This project contributes an adaptive, privacy-aware, quality-aware hierarchical face retrieval system for event photo discovery. It extends structured identity-code retrieval with multi-view selfie fusion, event-size-aware indexing, quality-based ranking, asynchronous processing, and explicit biometric privacy controls.

Prepared for project documentation. Implementation references: backend/app/hierarchy.py, backend/app/routers/events.py, backend/app/tasks.py, backend/app/routers/users.py, and frontend/src/app/events/[id]/selfie/page.tsx.